O-I-588c
APRIL 9, 1962

SUPERSEDING
Int. Fed. Spec. O–I–00588b(Army–QMC)
December 30, 1960 and
Fed. Spec. O–I–588a
February 3, 1954

FEDERAL SPECIFICATION

INSECTICIDE, WATER EMULSIFIABLE PETROLEUM OILS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The water emulsifiable petroleum oils covered by this specification are suitable for use as sprays for the control of insects on trees and shrubs (see 6.1).

1.2 Classification. The water emulsifiable petroleum oils shall be of the following types, as specified (see 6.2):

Type I—Dormant-plant spray (80 percent saturated petroleum oil).
Type II—Summer spray (90 percent saturated petroleum oil).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and Standards. The following specifications and standards, of the issues in effect on invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

TT–I–558—Ink, Marking Stencil, Opaque, for Nonporous Surfaces (Metals, Glass, etc.).
PPP–P–704—Pails: Shipping, Steel (1 Through 12 Gallon).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 791—Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–A–3941—Adhesive Paper Label, Water-Resistant.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

FSC 6840

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MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date for bids, shall apply.

Federal Insecticide, Fungicide, and Rodenticide Act.

(Copies may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

3. REQUIREMENTS

3.1 Composition. The water emulsifiable petroleum oils shall be liquids and shall contain the ingredients specified in table I for types I and II, as applicable.

TABLE I. *Composition of spray oils*

Ingredient	Requirement paragraph	Type I Percent by weight		Type II Percent by weight	
		Minimum	*Maximum*	*Minimum*	*Maximum*
Dormant-plant spray oil	3.1.1	95.0	....	....	....
Summer spray oil	3.1.1	....	....	95.0	....
Emulsifier	3.1.2	....	5.0	....	5.0

3.1.1 *Physical requirements.* The water emulsifiable petroleum oils shall conform to the physical requirements of table II for types I and II, as applicable, determined as specified in 4.3.

3.1.2 *Emulsifier.* The emulsifying agent shall be completely soluble in the oil, shall not be phytotoxic, and shall produce an emulsion with water that will not separate visible drops of oil on standing for 10 minutes at 80° ± 5°F. when tested as specified in 4.3.4.

3.2 Workmanship. The finished product shall be clean, free from undissolved material and shall conform to levels of quality established herein.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in

TABLE II. *Physical requirements of spray oils*

Property	Type I		Type II		Test method paragraph
	Minimum	Maximum	Minimum	Maximum	
Viscosity (Saybolt at 100° F.) seconds	90	120	50	80	4.3.1
Unsulfonated residue (percent by weight)	80	..	90	..	4.3.2
Distillation range: Distills up to 636° F., percent	10	25	..	..	4.3.3
Distillation range: Distills up to 520° F., percent	..	..	10	..	4.3.3
Distills up to 665° F., percent	..	..	80	..	4.3.3

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the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection for acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL–STD–105 except as otherwise indicated herein.

4.2.1 *Inspection of components and materials.* In accordance with 4.1 above, the supplier shall be responsible for insuring that the quality assurance provisions for components or materials shall be in accordance with this specification and with subsidiary specifications and drawings referenced herein to the extent applicable. In the event of conflict, this specification shall govern.

4.2.2 *Inspection of the end item.*

4.2.2.1 *Examination of the end item.* An examination shall be made of filled containers in accordance with the classification of defects set forth below. The sample unit shall be one filled container. The lot size shall be the number of containers of the same type and capacity submitted for acceptance at one time. The sample size for examination shall be at level L–8 with an acceptable quality level (AQL) of 1.0 expressed as defects per 100 units.

Examine	*Defect*
Container	Not type specified. Not interior coated (as applicable).
Workmanship	Any seepage or leakage. Any breaks or dents.
Label data markings	Omitted, incomplete, not clear, illegible or smeared. Not in accordance with specification (see 5.3).
Contents	Unclean, contains foreign material.

4.2.2.2 *Examination of net contents.* The sample unit shall be one filled container. The lot size shall be the number of containers of the same type and capacity submitted for acceptance at one time. The sample size shall be level L–3. The average net content per container shall be not less than specified.

4.2.2.3 *Examination of preparation for delivery.* An examination will be made to determine that markings, packaging, packing, contents and weights comply with the requirements of section 5 of this specification. Defects will be scored in accordance with the classification below. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closure defects listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of containers in the inspection lot fully prepared for delivery. The sample size shall be L–4 and the acceptable quality level (AQL) shall be 4.0, expressed as defects per 100 units (containers).

Examination	*Defect*
Markings (as applicable)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application.
Materials (as applicable)	Any nonconforming component including closures; component missing, damaged, or otherwise defective.
Workmanship (as applicable)	Inadequate application of components such as: Incomplete closure of case liners, container flaps, loose or inadequate sealing, strapping or stapling; bulged or distorted containers.
Contents or weight	Number per container is more or less than required.
Exterior and interior (as applicable)	Gross or net weight exceeds requirement.

4.2.3 *Testing of the end item.* The end item shall be tested for compliance with the characteristics specified in table III. The sample

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TABLE III. *End item testing requirements*

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Requirements applicable to individual unit	Results reported to nearest
Composition	3.1	Table II	1	X	percent
Viscosity	3.1.1 table II	4.3.1	1	X	second
Unsulfonated residue (percent by weight)	3.1.1 table II	4.3.2	1	X	percent
Distillation range	3.1.1 table II	4.3.3	1	X	percent
Emulsifier	3.1.2	4.3.4	1	X	10 seconds

unit shall be the quantity of material required to complete each test one time. The lot size shall be the number of containers of the end item submitted for acceptance at one time. The sample size for tests shall be at level L–2. There shall be no failure to meet any characteristic.

4.3 Test procedures.

4.3.1 *Viscosity (Saybolt).* The viscosity shall be determined in accordance with method 304.8 of Fed. Test Method Std. No. 791.

4.3.2 *Unsulfonated residue.* The unsulfonated residue shall be determined in accordance with method 5441.2 of Fed Test Method Std. No. 791.

4.3.3 *Distillation range.* The distillation range shall be determined in accordance with method 1002.3 of Fed. Test Method Std. No. 791.

4.3.4 *Emulsion stability.*

4.3.4.1 *Apparatus.* The test emulsion shall be prepared in a 250-ml. Griffin low-form beaker (Corning Cat. LP 28, No. 1000 or the equivalent). A stirrer shall be used which consists of a rod 10 mm. in diameter and 50 mm. long to which a 5 mm. diameter rod has been sealed or welded at right angles to make a T-shaped stirrer. The stirrer may be made of glass or metal. The stirrer shall be raised off the bottom of the beaker just enough to permit free rotation.

4.3.4.2 *Hard water.* The hard water specified in 4.3.4.3 shall have the following composition, and is designed to provide a hardness of 342 p.p.m. calculated as calcium carbonate:

$CaCl_2$ (anhydrous) .....0.3037 g.
$MgCl_2.6H_2O$ .....0.1388 g.
Distilled water to make ..1 liter

4.3.4.3 *Emulsification.* Place 99 ml. of hard water (4.3.4.2) at 80° ± 5°F. in the beaker and stir at 1000 ± 50 r.p.m. Add 1.0 ml. of the water emulsifiable oil from a pipette and continue the stirring for one minute. Pour the emulsion thus formed into a 100-ml. graduated glass-stoppered cylinder, note the time, and set aside for exactly 10 minutes.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civilian procurements.)

5.1 Packing. Packing shall be level A and B or C, as specified (see 6.2).

5.1.1 *Levels A and B.* Five gallons of one type insecticide shall be filled into a 5-gallon capacity pail, with interior coating, conforming to type I, class 3 of PPP–P–704. Closures shall be screw or snap-on type and shall be provided with a push-pull pouring spout. Each closure shall be equipped with a tamperproof export overseal with the Federal Identification Stock No. embossed on top.

5.1.2 *Level C.* Insecticides shall be furnished in shipping containers complying with

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the carrier rules and regulations applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 *Labeling.* The labeling shall be registered under the Federal Insecticide, Fungicide and Rodenticide Act of 1947 as amended, in compliance with its requirements. The label data shall be durably and legibly marked on the main panel of the body by lithographing, stenciling, or printing, in accordance with MIL–STD–129, with a yellow, noncorrosive, weather-resistant, waterproof ink meeting the requirements of TT–I–558. If the label data are too voluminous to be applied in this manner, instructions for use and caution statement may be placed on a label made of white paper as specified in MIL–STD–129. Labels shall be securely affixed in place with water-resistant label adhesive as specified in MIL–A–3941 and shall be waterproofed by coating the entire outer surface of the label with the same adhesive.

5.2.1.1 *Dormant-plant spray.* The dormant-plant spray, type I, label data shall be as follows:

INSECTICIDE, WATER EMULSIFIABLE PETROLEUM OIL, DORMANT-PLANT SPRAY

Stock No.....

Active ingredients	*Percent by weight*
Petroleum oil	.....
Inert ingredients ...	.....
Emulsifier	.....
Total	100

Date of pack (month and year) .....
Net contents .....

INSTRUCTIONS FOR USE AS A DORMANT-PLANT SPRAY

On Trees and Shrubs For Which Dormant-Plant Oil Sprays Are Used. Use this water emulsifiable oil as a dormant-plant spray to control scale insects, aphids, and spider mites including the San Jose scale, the oyster shell scale, the euonymus scale, the European red mite, and other species wintering on the trees in the egg stage.

For Evergreens, Deciduous Shrubs, and Trees. Dilute at the rate of 3 gallons (or parts by volume) of the oil to 97 gallons of water (or parts by volume) (contains about 3 percent oil). For small quantities, dilute one-half cupful with 1 gallon of water.

Mixing. Stir the oil rapidly and add water slowly until a fluid milky emulsion is formed, then add the remainder of the water rapidly.

CAUTION: Do not use this emulsifiable oil without diluting with water as directed. Also, do not apply the finished spray in freezing weather. It is preferable to apply it when the temperature is above 40°F. The safest period for application is in the spring just before the new growth starts.

5.2.1.2 *Summer spray.* The summer spray, type II, label data shall be as follows:

INSECTICIDE, WATER EMULSIFIABLE PETROLEUM OIL, SUMMER SPRAY

Stock No.....

Active ingredients	*Percent by weight*
Petroleum oil	.....
Inert ingredients ...	.....
Emulsifier	.....
Total	100

Date of pack (month and year) .....
Net contents .....

INSTRUCTIONS FOR USE AS A SUMMER SPRAY

On Trees, Shrubs, and Other Plants For Which Summer Oil Sprays Are Used. Use this water emulsifiable oil as a summer spray to control mites, the crawlers of scale insects, lacebugs, and mealybugs when applied to the foliage, especially on tender plants and shrubs.

For Tender Plants. Dilute at the rate of 1 gallon of emulsifiable oil in 99 gallons of water (contains about 1 percent oil). For small quantities dilute 2½ tablespoonfuls with 1 gallon of water. To avoid plant in-

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jury, wash or syringe the more tender plants with water an hour or so after treatment.

For More Hardy Shrubs and Trees. Dilute 2 gallons of the oil with 98 gallons of water (contains about 2 percent oil), or 5 tablespoonfuls with 1 gallon of water.

Extracts of pyrethrum, rotenone, or nicotine are sometimes added to increase the effectiveness of the spray against lacebugs and certain other sucking insects.

Mixing. Stir the oil rapidly and add water slowly until a fluid milky emulsion is formed, then add the remainder of the water rapidly.

CAUTION: Do not use this emulsifiable oil without diluting with water as directed.

5.2.2 *Civil agencies.* In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Fed. Std. No. 123.

5.2.3 *Military requirements.* In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use.

6.1.1 *Type I.* Dormant-plant spray oil covered by this specification is intended for use in the control of scale insects, aphids, and spider mites including the San Jose scale, the oyster shell scale, the euonymus scale, the European red mite and other species wintering on the trees in the egg state.

6.1.2 *Type II.* Summer spray oil covered by this specification is intended for use in the control of scale insects, lacebugs, and mealybugs when applied to the foliage, especially on tender plants and shrubs.

6.2 Ordering data. Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type required.
(*c*) Inspection responsibility, methods and test (see Section 4).
(*d*) Selection of applicable level of packing (see 5.1).

6.3 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:
Insecticides, oil emulsion, liquid.
Carload minimum weight 40,000 pounds.

Motor:
Insecticides, oil emulsion, liquid.
Motor volume minimum weight 40,000 pounds.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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